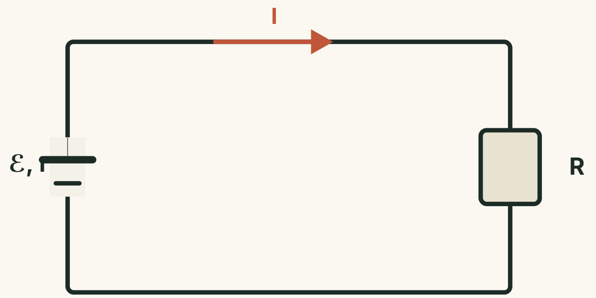
ELECTRIC CIRCUITS

Ohm's Law for Closed Circuits

The whole loop at once: a source pushes charge, the circuit spends the energy, and the battery warms.



OPENING QUESTION

Why does a battery heat up — and what is the expression for the heat it generates?

THE BIG PICTURE

A closed circuit has two parts.

The current is the same everywhere in a series loop — including inside the battery.

EXTERNAL CIRCUIT

Everything outside the source — resistance R, wires, switch — where the current does useful work.

INTERNAL CIRCUIT

The inside of the source itself, with EMF $\mathcal{E}$ and internal resistance r.

INSIDE THE BATTERY

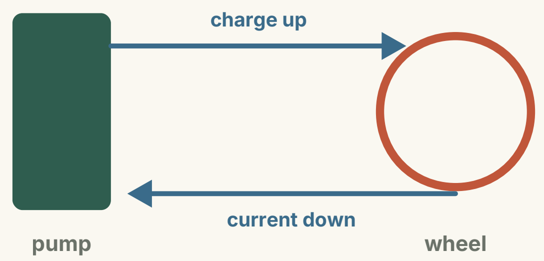
What does the source actually do?

A source does not create charge — it pumps charge that is already there.

Positive charge is pushed from the low-potential terminal back up to the high-potential terminal, **against** the electric force.

THE QUESTION

Something inside must do work on the charges. What is it?



NEW QUANTITY

Electromotive force (EMF).

The work done by non-electrostatic forces per unit charge moved through the source.

$$\mathcal{E} = \frac{W}{q}$$

01 · Physical meaning

How strongly a source converts other energy into electric potential energy.

02 · Unit

The volt (V) — same unit as voltage, but not the same quantity.

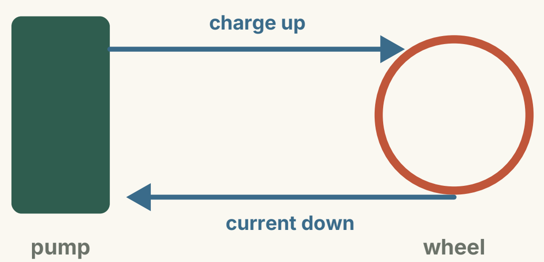
03 · Example

A 1.5 V cell gives every coulomb 1.5 J as it passes through.

ANALOGY

EMF and voltage: the water model.

The pump raises water against gravity — like the source raising charge to a higher potential.



EMF describes the pump's **lifting ability**; voltage describes the **drop** across each part of the path.

Water flowing back down through the wheel is the current doing work in the external circuit.

EXAMPLE 1

Example 1: understanding EMF.

Which statement about EMF is correct?

A $\mathcal{E} \propto$ work done by non-electrostatic forces, and $\propto$ 1/charge that passes

B EMF has the unit of voltage, so it is just the terminal voltage

C The more work the non-electrostatic forces do, the greater the EMF

D ✓ The EMF is set by the nature of the non-electrostatic forces in the source

ANSWER

D — $\mathcal{E} = W/q$ is a definition (a ratio), not a proportionality. Its value is fixed by the source, independent of W and q.

EXAMPLE 2

Example 2: comparing two batteries.

A lead–acid battery has $\mathcal{E}$ = 2 V; a dry cell, 1.5 V. Which is correct?

A The terminal voltage of the lead–acid battery is 2 V

B The dry cell converts 1.5 J of chemical energy per second

C The EMF of the lead–acid battery changes with the circuit

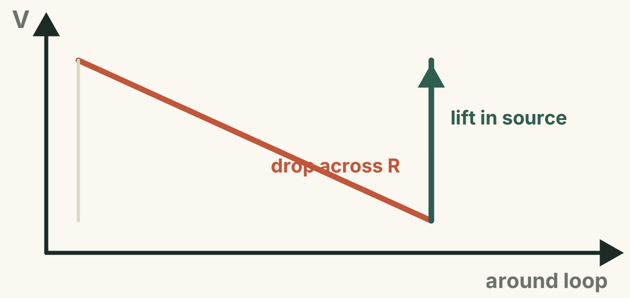
D ✓ For the same charge, non-electrostatic forces do more work in the lead–acid cell

ANSWER

D — $\mathcal{E} = W/q$, so a larger EMF means more work per unit charge. Terminal voltage $\leq \mathcal{E}$ and depends on the circuit; EMF does not.

How does the potential change around the loop?

The battery is the staircase of the circuit: everything that comes down must be carried back up.

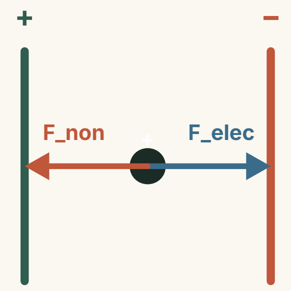


Around the external circuit the potential **falls** step by step as the current passes each resistance.

Inside the source it is lifted back up — otherwise the current could not keep flowing.

The non-electrostatic force.

Between the terminals the electric field pushes positive charge from + to –.



To move charge the other way, the source provides a **non-electrostatic force** that does positive work, lifting its potential energy.

In a battery this force is chemical; in a generator, electromagnetic.

ENERGY VIEW

Where the energy comes from.

The non-electrostatic force converts some other store of energy into electric potential energy.

CHEMICAL BATTERY

Chemical energy → electric potential energy, via chemical reactions at the electrodes.

GENERATOR

Mechanical energy → electric potential energy, via electromagnetic induction.

SUMMARY

EMF, in one box.

$$\mathcal{E} = \frac{W}{q}$$

THE ONE IDEA

EMF measures how strongly a source converts other energy into electric potential energy. Its value is decided by the source — not by W, q, or the external circuit.

CHECK YOURSELF

Check yourself.

Two quick questions on what EMF is — and is not.

Q1 ✓ The EMF equals the energy the source supplies per unit charge through the circuit

Q2 ✓ EMF reflects the source’s ability to convert other energy into electrical energy — it is a number, not a force

ANSWER

Both correct. EMF is energy per unit charge (a scalar in volts), not a force, and not the stored energy or the power.

PRACTICE

Practice: two batteries, two circuits.

Lead–acid (2 V, 0.1 A) and a dry cell (1.5 V, 0.2 A) each run for 20 s. How much chemical energy is consumed?

ENERGY CONVERTED BY THE SOURCE

$$W = \mathcal{E} I t$$

Pb $2 \times 0.1 \times 20 = 4$ J

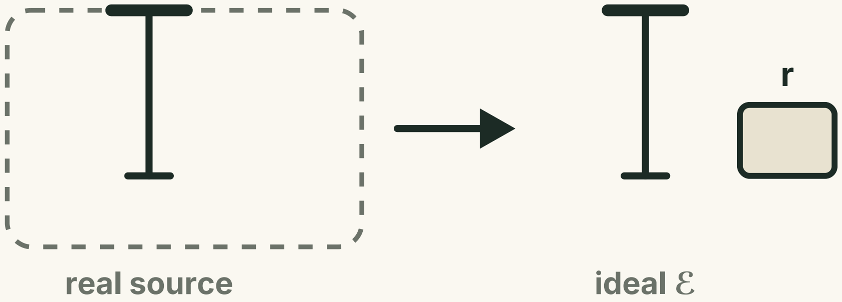
Dry $1.5 \times 0.2 \times 20 = 6$ J

READING IT

The dry cell converts more total energy here — but “converting ability” means energy per unit charge, which the EMF captures.

A real source = ideal source + resistor.

Electrolytes and electrodes resist current too. Pull that resistance out as an ordinary resistor r.



KEY RELATIONSHIP

Terminal voltage.

Energy per unit charge from the source = energy dropped on R + energy dropped on r.

$$\mathcal{E} = U_{ext} + Ir$$

AVAILABLE AT THE TERMINALS

$$U_{ext} = \mathcal{E} - Ir$$

CURRENT IN THE LOOP

$$I = \frac{\mathcal{E}}{R + r}$$

More current → more voltage lost inside. With I = 0, U = $\mathcal{E}$.

EXAMPLE 3

Example 3: terminal voltage.

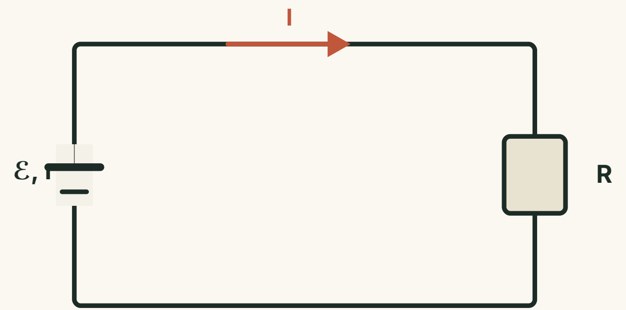
R = 2.0 Ω, ℰ = 3.0 V, r = 1.0 Ω. After the switch closes, what is the terminal voltage?

1. **Loop current**

$$I = \frac{\mathcal{E}}{R + r} = \frac{3.0}{2.0 + 1.0} = 1.0\ \mathrm{A}$$

2. **Terminal voltage**

$$U = \mathcal{E} - Ir = 3.0 - 1.0 \times 1.0 = 2.0\ \mathrm{V}$$



EXAMPLE 4

Example 4: find $\mathcal{E}$ and r from meter readings.

Closed: 1.6 V, 0.4 A. Opening S changes the readings by 0.1 V and 0.1 A → (1.7 V, 0.3 A).

1. **Internal resistance**

$$r = \frac{\Delta U}{\Delta I} = \frac{0.1}{0.1} = 1\ \Omega$$

2. **EMF from either state**

$$\mathcal{E} = 1.6 + 0.4 \times 1 = 2.0\ \mathrm{V}$$

THE TRICK

Two (U, I) states and $\mathcal{E}$ = U + Ir give two equations; subtract to get r = ΔU/ΔI.

EXAMPLE 5

Example 5: a circuit with a motor (part 1).

$\mathcal{E}$ = 6 V; bulb L rated "3 V, 0.9 W"; rheostat R = 8 Ω; motor coil R_0 = 2 Ω. Find the internal resistance.

1. **Bulb at normal operation**

$$R_L = \frac{U^2}{P} = \frac{3^2}{0.9} = 10\ \Omega, \quad I = \frac{P}{U} = 0.3\ \text{A}$$

2. **Apply $\mathcal{E}$ = I(R_L + r + R)**

$$6 = 0.3(10 + r + 8) \ \Rightarrow\ r = 2\ \Omega$$

SET-UP

With S at position 1 only the bulb runs, so the same current flows through bulb, rheostat and source.

Example 5: power and efficiency (part 2).

With S at 2 the bulb still works normally, so $I = 0.3$ A. The motor's voltage: $U_M = \mathcal{E} - I(R_L + r) = 2.4$ V.

P_in $U_M I = 0.72$ W

P_heat $I^2 R_0 = 0.18$ W

P_out $P_{in} - P_{heat} = 0.54$ W

η $P_{out} / P_{in} = 75\%$

WATCH OUT

A motor is not a pure resistor — Ohm's law applies to its coil resistance only, not to the whole device.

APPLICATION

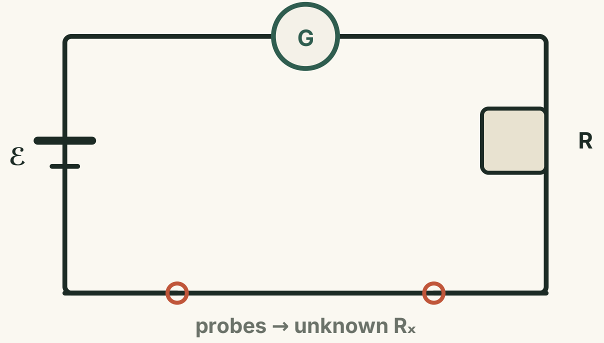
The ohmmeter.

A battery ($\mathcal{E}$, r), a galvanometer G and an adjustable R in series — Ohm's law for closed circuits, used to measure.

A known EMF drives a current that depends on the unknown resistance across the probes.

ZERO ADJUSTMENT

Touch the probes together and tune R until the pointer reads full scale ("0 Ω").



HOW THE SCALE WORKS

From current to resistance.

Each unknown R_x gives exactly one current — so the current scale can be relabelled directly in ohms.

LOOP CURRENT

$$I = \frac{\mathcal{E}}{R + R_g + r + R_x}$$

SOLVE FOR THE UNKNOWN

$$R_x = \frac{\mathcal{E}}{I} - (R + R_g + r)$$

Internal resistance R_int = R + R_g + r.

WHY THE SCALE IS REVERSED

Probes together → full deflection = “0 Ω”; probes apart → no deflection = “∞”. Larger R_x means smaller current.

WORKED EXAMPLE

Worked example: the ohmmeter scale.

Full-scale current I_g = 500 μA; cell EMF 1.5 V. Find the internal resistance and label the scale.

ZEROING (PROBES TOGETHER)

$$R_{int} = \frac{\mathcal{E}}{I_g} = \frac{1.5}{500\ \mu\text{A}} = 3000\ \Omega$$

200 μA $R_x = 4.5 \times 10^3\ \Omega$

300 μA $R_x = 2 \times 10^3\ \Omega$

500 μA $R_x = 0\ \Omega$ (full scale)

MEASURING

$R_x = \mathcal{E}/I - R_{int}$. Each current mark corresponds to exactly one resistance.

The ohmmeter scale (continued).

Total internal resistance is 3000 Ω. At ⅓ of full scale, what resistance is being measured?

1. **Current is I_g / 3**
   so the loop resistance is 3× the internal value

2. **Unknown resistance**
   R_x = 3 × 3000 − 3000 = 6000 Ω

HANDY RULE — AT 1/N OF FULL SCALE

$$R_x = (n-1)R_{int}$$

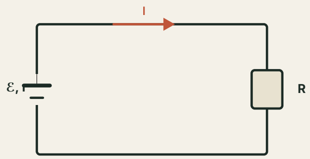
The scale is non-linear: the centre reads R_int, and it crowds toward ∞.

SUMMARY

One equation holds the whole lesson: $\mathcal{E} = U_{ext} + Ir$.

AND THE BATTERY HEATS AT A RATE

$$P = I^2 r$$



The source gives every unit of charge an energy $\mathcal{E}$; part is delivered outside, part is lost inside.

That inside part, $P = I^2 r$, is exactly why the battery warms up.